Hyperbolic Functions: Reciprocal Functions

Questions 1–28

Question 1. Write down the exponential form of each of these, giving your answer:

a) $\operatorname{cosech} x$
b) $\operatorname{sech} x$
c) $\coth x$

Question 2. Write down the exact value of each of these in terms of exponentials, and then to 3 significant figures:

a) $\operatorname{cosech} 1$
b) $\operatorname{sech} 1$
c) $\operatorname{cosech}(-1)$
d) $\operatorname{sech}(-1)$

Question 3. Show that $\operatorname{cosech}(-x) = -\operatorname{cosech} x$.

Question 4. Show that $\operatorname{sech}(-x) = \operatorname{sech} x$.

Question 5. By using the exponential forms of $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$, find a simplified exact value for each of these:

a) $\operatorname{cosech}(\ln 3)$
b) $\operatorname{sech}(\ln 6)$
c) $\operatorname{cosech}(\ln \sqrt{5})$
d) $\operatorname{sech}(4 \ln 3)$

Question 6. Show that $\operatorname{cosech}(\ln k) = \dfrac{2k}{k^2 - 1}$ where k is a positive constant.

Question 7. Show that $\operatorname{sech}(3\ln k) = \dfrac{2k^3}{k^6+1}$ where k is a positive constant.

Question 8. It is given that $f(x) = \operatorname{cosech}(2\ln x) - \operatorname{sech}(2\ln x)$ where $x > 0$ and $x \neq k$ for some constant k.

a) Explain why $k = 1$.

b) Show that $f(x) = \dfrac{4x^2}{(x^4-1)(x^4+1)}$.

Question 9. By using the exponential forms of $\sinh x$ and $\cosh x$, show that $\coth x = \dfrac{e^{2x}+1}{(e^x+1)(e^x-1)}$.

Question 10. By using the exponential forms of $\operatorname{cosech} x$, $\operatorname{sech} x$ and $\coth x$, find a simplified exact value for each of these:

a) $\coth(\ln 2)$
b) $\coth(\ln \sqrt{2})$
c) $\coth\left(\frac{1}{4}\ln 8\right)$

Question 11. Show that $\coth(-x) = -\coth x$.

Question 12. Find a simplified exact value for each of these:

a) $\coth(-\ln 3)$

b) $\coth(-5\ln 2)$

Question 13. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech} x = 1$

b) $\operatorname{cosech} x = 2$

c) $\operatorname{cosech} x = -\frac{1}{5}$

Question 14. Solve the following equations, writing your answers in exact form:

a) $\operatorname{sech} x = 1$

b) $\operatorname{sech} x = \frac{1}{2}$

c) $\operatorname{sech} x = \frac{2}{3}$

Question 15. Solve the following equations, writing your answers in exact form:

a) $\coth x = 2$

b) $\coth x = \frac{3}{2}$

c) $\coth x = -\frac{5}{3}$

Question 16. Solve the following equations, writing your answers in exact form:

a) $5\,\mathrm{sech}\, x - 2\tanh x = 5$

b) $8\,\mathrm{cosech}\, x - 3\coth x = 27$

c) $2\tanh x - 5\coth x = 9$

Question 17. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech}^2 x = 9$

b) $\operatorname{sech}^2 x = \frac{1}{16}$

c) $\coth^2 x = 3$

Question 18. Solve the following equations, writing your answers in exact form:

a) $\operatorname{cosech}^2 x + \operatorname{cosech} x - 20 = 0$

b) $10 \operatorname{sech}^2 x - 9 \operatorname{sech} x + 2 = 0$

c) $3 \coth^2 x - 16 \coth x - 64 = 0$

Question 19. For each of the following functions, sketch the graph and state whether it is one-to-one or if it is many-to-one:

a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

Use set notation for the following questions which require a range of values.

Question 20. State the largest possible domain and range of the following functions:

a) $y = \operatorname{cosech} x$
b) $y = \operatorname{sech} x$
c) $y = \coth x$

Question 21. State the range of the following functions given $x > 0$:

a) $y = \operatorname{cosech} x$

b) $y = \operatorname{sech} x$

c) $y = \coth x$

Question 22. Sketch each of the following functions, stating their largest possible domain and range:

a) $f(x) = \operatorname{cosech}(x + 3)$

b) $g(x) = 2 + \operatorname{sech} x$

c) $h(x) = \coth(x - 1) - 1$

Question 23. State the largest possible domain and range for each of the following functions:

a) $f(x) = \operatorname{cosech}(x-2)+2$
b) $y = 2 - \operatorname{cosech} x$
c) $g(x) = 3\operatorname{sech}(x+3)$
d) $y = 4 + 3\operatorname{sech} x$
e) $h(x) = \coth(x+4) - 3$
f) $y = 2 - \frac{1}{3}\coth x$

Question 24. State the largest possible domain and range for each of the following functions:

a) $y = \operatorname{cosech}^2 x$
b) $y = \operatorname{sech}^2 x$
c) $y = \coth^2 x$

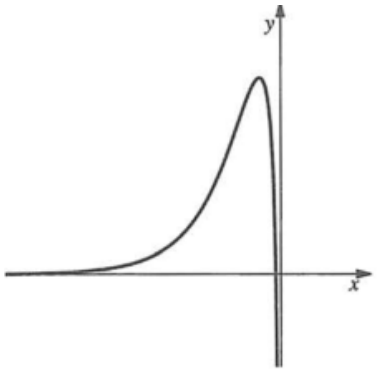
Question 25.

a) Write $\operatorname{cosech} x + \operatorname{sech} x$ in the form $\dfrac{ae^{bx}}{e^{cx} - 1}$ where a, b and c are real constants.

b) Hence explain why, for $x > 0$, $\operatorname{cosech} x + \operatorname{sech} x > \dfrac{a}{e^{cx} - 1}$.

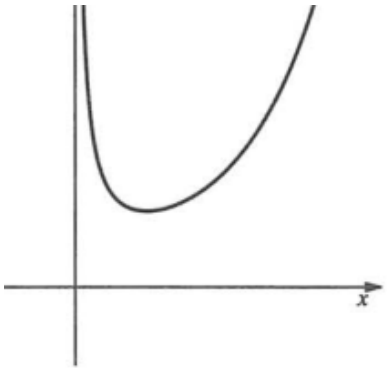
Question 26. Show clearly that, for $x > 0$, $\coth x - \tanh x > 4e^{-2x}$.

Question 27. The graph below shows $f(x) = \operatorname{cosech} x + 8 \operatorname{sech} x$ where $x < 0$.



Find the exact largest possible range of f, writing your answer in set notation.

Question 28. The graph below shows $f(x) = \cosh x + \coth x$ where $x > 0$.



Find the exact largest possible range of f, writing your answer in interval notation.